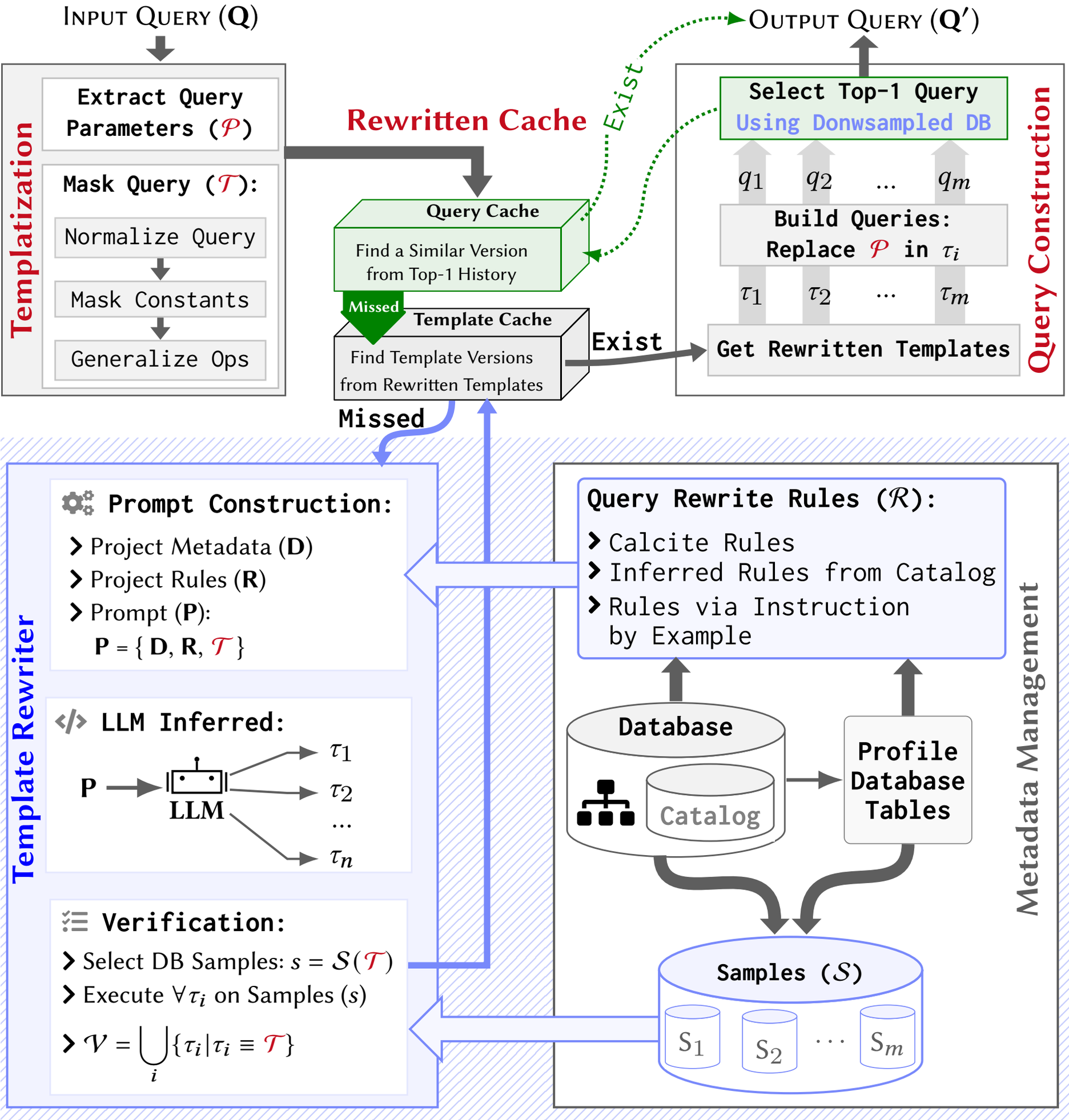

Input Query (Q)
Templatization
Extract Query Parameters ($\mathcal{P}$)
Mask Query ($\mathcal{T}$):
Normalize Query
Mask Constants
Generalize Ops
Rewritten Cache
Query Cache
Find a Similar Version from Top-1 History
Missed
Template Cache
Find Template Versions from Rewritten Templates
Exist
Missed
Exist
Output Query (Q')
Select Top-1 Query Using Donwsampled DB
q_1 q_2 ... q_m
Build Queries: Replace $\mathcal{P}$ in τ_i
τ_1 τ_2 ... τ_m
Get Rewritten Templates
Query Construction
Template Rewriter
Prompt Construction:
Project Metadata (D)
Project Rules (R)
Prompt (P):
P = { D, R, $\mathcal{T}$ }
LLM Inferred:
P
LLM
τ_1
τ_2
...
τ_n
Verification:
Select DB Samples: $s = \mathcal{S}(\mathcal{T})$
Execute $\forall \tau_i$ on Samples (s)
$\mathcal{V} = \bigcup_i \{\tau_i | \tau_i \equiv \mathcal{T}\}$
Query Rewrite Rules ($\mathcal{R}$):
Calcite Rules
Inferred Rules from Catalog
Rules via Instruction by Example
Database
Catalog
Profile Database Tables
Samples ($\mathcal{S}$)
S_1 S_2 ... S_m
Metadata Management